# Analysis Approaches for the Biomarker-by-Drug Interaction in Aggregated N-of-1 Trials: A Catalog

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**Purpose.** This paper lists every analysis approach for the biomarker-by-drug interaction that has been used, evaluated or proposed in this project. It covers:

- the published analysis of Hendrickson et al. (`orig`, commit `58b32a9`) and its options;
- the nine specifications of manuscript 02;
- the test procedures of manuscripts 08 and 10, and the analysis strategies of manuscript 06;
- the summary-measures statistics of `docs/37`;
- the alternatives discussed in `docs/40`, `docs/41` and `docs/42`;
- the analyses in the covariance study (`08-covariance-study.R`), including its follow-up run;
- the 14 analyses of the stress test (`docs/46`, `11-strawman-stress-test.R`), added on 2026-10-07 (Section 3.9).

The approaches are organized along the dimensions on which an analysis can vary (Section 2). The named approaches are then listed with their source, their status, and what is known about the size of their test (Sections 3 and 4). Section 5 summarizes which approaches are known to hold their size, and Section 6 lists the gaps.

**Status note (2026-10-07).** The catalog was written on 2026-10-04, while the covariance-study follow-up was running. Its status entries have been updated with dated notes: the follow-up is reported in `docs/45`, and the stress test of `docs/46` has since evaluated 14 analyses in 41 cells. Two statements of the original are corrected where they occur: the unstructured covariance with Kenward-Roger is not correctly sized everywhere, and the flexible-time size excess is confined to model-based and Kenward-Roger inference.

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## Notation index and glossary

Notation follows `analysis/report/NOTATION.md`. Symbols marked *local* are defined here.

Table 1: Notation index and glossary of symbols used in this document

| Symbol | Meaning | Status |
|---|---|---|
| $\beta_{bm:D}$ | biomarker-by-drug interaction, the estimand | canonical |
| $D_{it}$, $D_{bc,it}$ | binary drug state (`Db`); exposure-decayed drug indicator (`Dbc`) | canonical |
| $\lambda$, $t_{sd}$, $t_{1/2}$ | decay rate; time since discontinuation; carryover half-life | canonical |
| $t^a_{1/2}$ | half-life assumed by the analysis (`docs/45`) | local |
| $L_{it}$ | lagged just-off-drug indicator (E3) | local |
| $\eta$ | measured expectancy covariate (manuscript 06) | canonical |
| $c_{bm}$ | biomarker-BR coupling in the stress-test constructs | canonical |
| $\beta_B$ | biomarker main effect (`docs/41`) | local |
| $\kappa$ | mean model-based SE over empirical SD | canonical |
| $\rho$ | AR(1) serial correlation | canonical |
| $N$ | total sample size across paths | canonical |

**Identifiers.** The IDs of Section 3 are local to this catalog. The stress test of `docs/46` uses its own IDs (S, A, C1, C2, DS, DA, ES, EA, F1, F2, R, G1, G2, F1A), several of which collide with IDs here (C1, C2, F1, F2, G1, G2). Following the NOTATION convention for paper-local labels, they are cited here with a prefix: 46-F1 is the stress test's F1, and F1 alone is the covariance-study follow-up arm of Section 3.8.

- $\beta_{bm:D}$**.** The biomarker-by-drug interaction, the estimand.
- **`Db, Dbc`.** Binary drug state, $D_{it}$; exposure-decayed drug indicator, $D_{bc,it}$ (1 on drug, $e^{-\lambda t_{sd}}$ off drug).
- **Size.** Type I error of the interaction test; nominal 0.05. An approach is *correctly sized* in a cell if its null rejection rate is within about 2 Monte Carlo standard errors of 0.05.
- $\kappa$**.** Mean model-based standard error over the empirical standard deviation of the estimate. Below 1 means anticonservative.
- **CS, AR(1).** Compound symmetry; first-order autoregressive correlation.
- **CR2.** Bias-reduced cluster-robust (sandwich) standard error, with Satterthwaite (Bell-McCaffrey) degrees of freedom.
- **KR.** Kenward-Roger standard-error and degrees-of-freedom correction.
- **MD.** Mancl-DeRouen bias-corrected sandwich for GEE.

- **cLDA.** Constrained longitudinal data analysis: baseline kept in the response with a common mean.
- **Construct.** The data-generating process a result was obtained under. *Published CS* is the 58b32a9 construct (`docs/36`); *configuration C* is separable AR(1) with graded coupling; *package* is the current package construct used by manuscripts 02, 06, 08 and 10.

## 1. Summary

1. **An analysis is a combination of about a dozen choices.** They are: the estimator family, the exposure coding, the time adjustment, the expectancy term, the baseline handling, the random effects, the working covariance, the standard error and degrees of freedom, the path term, the biomarker coding, the target coefficient, and the design-specific branches (Section 2).
2. **About 60 distinct approaches** have been used, evaluated or proposed (Section 3). Most vary one or two choices against a reference.
3. **Correct size depends mainly on the working covariance and the small-sample correction**, not on the exposure coding, the time adjustment or the baseline coding. Each of those moves the Type I error by at most about 0.02. The covariance and its correction move it by up to 0.10 (Section 4).
4. **Known to hold their size in every construct tested:**

   - an unstructured covariance with Kenward-Roger, baseline as a row (covariance study: 0.046 to 0.072);
   - CR2 standard errors with a random intercept and `corCAR1` (papers 02 and 10: 0.045 to 0.049);
   - GEE with the Mancl-DeRouen correction (papers 08 and 10).

   **Known to fail somewhere:**

   - the published random intercept under AR(1) data (Type I error 0.14 to 0.16 in CO);
   - any unstructured covariance with Satterthwaite only (0.072 to 0.104);
   - naive GEE (1.5 to 1.9 times nominal);
   - flexible time adjustments in Hybrid under carryover (about 0.07, under every covariance tested).

   *Corrected (2026-10-07).* The unstructured covariance with Kenward-Roger is slightly liberal over the eight covariance-study null cells (mean 0.058; `docs/45`, Section 4.1) and fails the stress test of `docs/46`, where it is 46-A: pooled null rejection 0.057, 0.087 at $N = 35$, and 0.195 to 0.416 in the three original prognostic cells. The flexible-time excess is real under model-based and Kenward-Roger inference but absent under CR2 (`docs/45`, Section 4.4; `docs/42`, update after Section 1).

5. **Open:** the Kenward-Roger and CR2 versions of the baseline-covariate coding, CR2 with structured covariances under AR(1) data, and the mechanism behind the row coding's power under AR(1). All are in the follow-up run now in progress (Section 6). *Answered (2026-10-07)* by `docs/45`: the covariate coding is slightly liberal with Kenward-Roger (0.059) and nominal with `cs` and CR2 (0.048); CR2 with a parsimonious covariance is nominal in every construct (0.044 to 0.054); the row coding's power under AR(1) comes from the shared biomarker effect $\beta_B$ (Section 3.8).
6. **The stress test (2026-10-07; `docs/46`, Section 3.9).** 14 analyses, 41 cells, 28,000 trials. Ten survive the size and sensitivity screens: 46-S, 46-C1, 46-C2, 46-DS, 46-ES, 46-EA, 46-F1, 46-F2, 46-G1 and 46-G2. Five of them are also nominal in every prognostic cell: 46-ES, 46-EA, 46-F1, 46-G1 and 46-G2. Among these, 46-G2 and 46-F1 tie on maximin shortfall (0.232 and 0.242), and the rule selects 46-F1, which has fewer covariance parameters: a random intercept with `corCAR1` residuals, CR2 inference, and the expectancy term `De` with `bmc:De` (`docs/49`).

## 2. Dimensions of an analysis

Each dimension lists its options, where each was used or proposed, and its current status. "Evaluated" means a simulation result exists; the source column gives the paper.

### 2.1 Estimator family

Table 2: Estimator family options with source and evaluation status

| Option | Source | Status |
|---|---|---|
| Linear mixed model, conditional (`lmer`, `nlme::lme`) | orig, 02, 06, 08, 10, covariance study | evaluated |
| Marginal model with structured covariance (MMRM, `mmrm`, `gls`) | covariance study; `docs/41`, `docs/42` | evaluated |
| Summary measures: per-participant on-minus-off contrast, regressed on the biomarker (E9) | `docs/37`; 02 (G5); 08 (continuous-biomarker contrast) | evaluated |
| Strict repeated-measures ANOVA, dichotomized biomarker | `docs/26`; 08 | evaluated (08) |
| GEE with sandwich variance | 08; 10 | evaluated |
| Two-stage approaches other than E9 | 08 (literature review) | not evaluated |

| Option | Source | Status |
|---|---|---|
| Nonlinear component-matched model (Gompertz BR, PB, TV) | `docs/31`; 06 | partly evaluated (06, recovery study) |
| Bayesian hierarchical N-of-1 model | 02 (named as out of scope) | not evaluated |
| Randomization or permutation test | 08 (literature) | not evaluated |
| Latent-class mixture model | manuscript 03 | evaluated (different estimand) |

### 2.2 Drug-exposure coding

Table 3: Drug-exposure coding options E1 to E7 with source and status

| Option | Code | Source | Status |
|---|---|---|---|
| Binary on-drug indicator `Db` | E1 | orig; 02 G1 | evaluated |
| Exposure-decayed `Dbc`, assumed half-life | E2 | 02 G3; implementations/original | evaluated |
| Binary plus lagged just-off-drug term $L_{it}$ | E3 | 02 G2 (Jones-Kenward) | evaluated: no gain over E1 |
| Binary plus linear time-since-discontinuation | E4 | implementations/original (`simplecarryover`) | retired in 02 |
| Lag term crossed with biomarker ($L$, bm:$L$) | E5 | 02, earlier S7 | retired: worse than E1 |
| $t_{sd}$ crossed with biomarker | E6 | 02, earlier S7 | retired: worse than E1 |
| `Dbc` with half-life chosen by AIC | E7 | 02 G4; stress test (46-DS, 46-DA, 46-F1A) | evaluated: nominal in one cell (02); in the stress test 46-DS passes, 46-DA and 46-F1A fail size (pooled 0.065 and 0.057) |
| `Dbc` with alternative decay forms (Weibull, linear, power) | | 02 sensitivity blocks; NOTATION | evaluated as misspecification |

### 2.3 Time adjustment

Table 4: Time adjustment options with source and evaluation status

| Option | Source | Status |
|---|---|---|
| None | E9 (`docs/37`) | evaluated |
| Linear `t` | orig; 02; all mixed models above | evaluated |
| Linear `t` plus expectancy `De` | orig option (`useDE`); `docs/40`; stress test (46-F1, with `bmc:De`) | evaluated (2026-10-07): 46-F1 nominal and prognostic-robust; the selected analysis |
| Natural spline in `t`, 3 df | `docs/40`, `docs/42`; covariance study | evaluated: size excess in the Hybrid carryover cell under model-based and Kenward-Roger inference, none under CR2 |
| Visit as factor | `docs/40`, `docs/42`; covariance study | evaluated: as the spline |
| Biomarker by linear time (`bmc:t`) | stress test (46-F2) | evaluated (2026-10-07): nominal, but fails the expectancy-prognostic cell |
| Phase indicators, phase by drug (attribution) | 06 (phase-augmented) | evaluated (06): no bias reduction, less power |
| Random slope on `t` | orig option (`t_random_slope`) | not evaluated |
| Path-specific intercepts (E9 stage 2) | `docs/37` (stratified E9) | closed form only |

### 2.4 Expectancy

Table 5: Expectancy modeling options with source and evaluation status

| Option | Source | Status |
|---|---|---|
| Not modeled | orig as published | evaluated |
| Design expectancy `De` as a fixed effect | orig option | population-mean limit (`docs/40`); evaluated with `bmc:De` in the stress test (46-F1, 46-G1, 46-G2, 46-F1A; 2026-10-07) |
| Absorbed by visit effects | `docs/40`, `docs/42` | evaluated |
| Measured belief covariate $\eta$ | 06 (balanced-placebo design) | evaluated (06) |

### 2.5 Baseline handling

Table 6: Baseline handling options with source and evaluation status

| Option | Source | Status |
|---|---|---|
| Ninth response row, coded off drug | orig | evaluated |
| Excluded, no covariate | 02 (all nine) | evaluated (02) |
| Covariate (ANCOVA) | `docs/41`; covariance study | evaluated with `cs` and `us`: Satterthwaite (main run), Kenward-Roger and CR2 (follow-up, `docs/45`) |
| cLDA: row with common baseline mean, unstructured covariance | `docs/41` | equivalent to the covariate coding only with $\beta_B$ freed at baseline (`docs/45`, Section 6) |
| Change from baseline | `docs/41` | identical to ANCOVA in the construct |

| Option | Source | Status |
|---|---|---|
| Baseline-by-drug moderator | `docs/41` | not evaluated (zero in the construct) |
| Row, with a separate biomarker effect at baseline | follow-up run (`us-KR-sepbm`); stress test (46-ES, 46-EA) | evaluated (2026-10-07): 46-ES and 46-EA nominal and prognostic-robust |

### 2.6 Random effects

Table 7: Random-effects structures with source and evaluation status

| Option | Source | Status |
|---|---|---|
| None (marginal covariance only) | MMRM; 10 (M0, M2) | evaluated |
| Random intercept | orig; 02; 10 (M1, M3) | evaluated |
| Random intercept and slope on `t` | orig option | not evaluated |
| Random intercept, post-baseline shift and expectancy coefficient, phase-specific residual variance | this discussion (implied by the published construct) | partly evaluated (2026-10-07): 46-G1 and 46-G2 have a fixed expectancy term and phase-specific residual variances, without the random shift or random expectancy coefficient |

### 2.7 Working covariance of the residuals

Table 8: Working residual covariance options with source and evaluation status

| Option | Source | Status |
|---|---|---|
| Independent | orig; 10 (M1) | evaluated |
| `corCAR1` (continuous-time AR(1)) | 02; 06; 08; 10 (M2, M3) | evaluated |
| Compound symmetry `cs` | covariance study (equals orig's random intercept) | evaluated |
| Heterogeneous CS `csh` | covariance study | evaluated |
| Unstructured `us` | covariance study; 10 | evaluated |
| Spatial exponential `sp_exp` | covariance study | evaluated |
| AR(1), Toeplitz, ante-dependence | `mmrm` options | not evaluated (assume equal spacing) |
| Phase-specific variance (`varIdent`) | this discussion; stress test | evaluated (2026-10-07): 46-C1, 46-C2, 46-G1, 46-G2 |
| GEE working correlation: independence, exchangeable, AR(1) | 08; 10 (AR(1)) | evaluated |

### 2.8 Standard error and degrees of freedom

Table 9: Standard error and degrees-of-freedom options with source and status

| Option | Source | Status |
|---|---|---|
| Model-based SE, Satterthwaite df (`lmerTest`) | orig | evaluated |
| Model-based SE, containment df (`nlme`) | 02; 06; 08 | evaluated |
| Model-based SE, Satterthwaite df (`mmrm`) | covariance study | evaluated |
| Kenward-Roger | covariance study (`us`); 10 | evaluated for `us`, `cs`, `csh` and the baseline covariate (`docs/45`): repairs `us` only |

| Option | Source | Status |
|---|---|---|
| CR2 sandwich, Satterthwaite df (`clubSandwich`) | 02 (G6 to G9); 10; stress test | evaluated; the inference of 46-S, 46-C1, 46-DS, 46-ES, 46-F1, 46-F2, 46-G1 and 46-F1A |
| CR2 sandwich, Satterthwaite df (`mmrm`) | follow-up run | evaluated (`docs/45`): nominal with parsimonious covariances, liberal with `us` (0.086) |
| CR0, CR1, CR3 sandwiches | 10 (discussed) | not evaluated |
| Naive GEE sandwich | 08; 10 | evaluated |
| Mancl-DeRouen GEE sandwich | 08; 10 | evaluated |
| OLS t, N − 2 df | E9 | evaluated |

### 2.9 Path or sequence term

Table 10: Path or sequence term options with source and evaluation status

| Option | Source | Status |
|---|---|---|
| None | orig and most | evaluated |
| Fixed path effect | `docs/37`, single-path study | evaluated: no effect on `lmer` power |
| Path-specific intercepts in E9 | `docs/37` | closed form only |

### 2.10 Biomarker coding and target

Table 11: Biomarker coding and target coefficient options with source and status

| Option | Source | Status |
|---|---|---|
| Uncentered biomarker | orig; 02 | evaluated |
| Centered biomarker | `docs/40`; covariance study | evaluated (target unchanged) |

| Option | Source | Status |
|---|---|---|
| Dichotomized biomarker | 08 (strict RM-ANOVA) | evaluated: costs 0.08 to 0.21 power |
| Target `bm:Db` | orig; 02 G1, G2 | evaluated |
| Target `bm:Dbc` | 02 G3, G4 | evaluated |
| Component-specific slopes | 06 | evaluated |
| Joint 2-df test (E5, E6) | 02 | retired |

### 2.11 Design-specific branches

Table 12: Design-specific analysis branch for the OL design

| Option | Source | Status |
|---|---|---|
| Biomarker-by-time test when no participant varies in drug state (OL design): `Sx ~ bm + t + bm*t` | orig | evaluated in the published results; confounded with natural course and expectancy |

## 3. Named approaches

Each row is a complete specification that has been fitted or proposed. Status is *evaluated*, *running* or *proposed*; the evidence column gives the construct and the key size result where one exists.

### 3.1 The published analysis and its options (orig, 58b32a9)

Table 13: Published analysis O1 and its options O2 to O5, with evidence on size

| ID | Specification | Status | Evidence on size |
|---|---|---|---|
| O1 | `lmer`, fixed part `Sx ~ bm + Db + t + bm*Db`, random intercept by `ptID`, baseline row, Satterthwaite (published default) | evaluated | published CS: 0.049 to 0.058 (Hybrid), 0.04 to 0.05 (CO). Published results: 0.01 to 0.09. AR(1) data: 0.14 to 0.23 in CO (`docs/36`) |

| ID | Specification | Status | Evidence on size |
|---|---|---|---|
| O2 | O1 plus `De` (`useDE = TRUE`) | proposed | dropped by Hendrickson et al. for collinearity; correlation with `Db` 0.59 (`docs/40`). Not simulated as specified; a CR2 analogue with `bmc:De` is 46-F1 |
| O3 | O1 with a random intercept and random slope on `t` by `ptID` (`t_random_slope = TRUE`) | proposed | |
| O4 | OL branch: fixed part `Sx ~ bm + t + bm*t`, random intercept by `ptID` | evaluated | confounded test; not a drug-by-biomarker interaction |
| O5 | Later vendored implementation: `nlme::lme(Sx ~ bm + t + Dbc + bm:Dbc)`, random intercept, `corCAR1` | evaluated as 02 G3 | see 02 G3 |

### 3.2 Manuscript 02 (carryover specifications; package construct)

All use `nlme::lme` with a random intercept and `corCAR1`, no baseline row, linear `t`.

Table 14: Manuscript 02 carryover specifications G1 to G9 with evidence on size

| ID | Specification | Status | Evidence on size |
|---|---|---|---|
| G1 | `Db`, target `bm:Db` | evaluated | mildly conservative (mean null 0.036 to 0.039) |
| G2 | G1 plus lag $L$ | evaluated | as G1; no power gain over G1 |
| G3 | `Dbc`, target `bm:Dbc` | evaluated | as G1 |
| G4 | G3, half-life chosen by AIC | evaluated (one cell) | nominal despite selection |
| G5 | Paired difference (E9) | evaluated | |
| G6 to G9 | G1, G2, G3, G4 with CR2 | evaluated | CR2 restores nominal size (0.045 to 0.049 in 10) |

### 3.3 Manuscripts 08 and 10 (test procedures and calibration; package construct)

Table 15: Manuscript 08 and 10 test procedures with evidence on size

| ID | Specification | Status | Evidence on size |
|---|---|---|---|
| T1 | Strict RM-ANOVA, dichotomized biomarker | evaluated (08) | dichotomization costs 0.08 to 0.21 power |
| T2 | Continuous-biomarker within-participant contrast | evaluated (08) | calibrated |
| T3 | `lme`, random intercept, `corCAR1` (production) | evaluated (08, 10 M3) | conservative, 0.029 to 0.033 (10) |
| T4 | GEE, naive sandwich | evaluated (08, 10) | anticonservative, 1.5 to 1.9 times nominal (08); 0.067 to 0.069 (10) |
| T5 | GEE, Mancl-DeRouen | evaluated (08, 10) | restores nominal size |
| M0 | OLS, no within-participant correlation | evaluated (10) | not anticonservative |
| M1 | Random intercept only | evaluated (10) | well calibrated in 10's construct |
| M2 | `corCAR1` only | evaluated (10) | intermediate |
| M3 | Random intercept plus `corCAR1` | evaluated (10) | conservative; CR2 fixes it; KR inert on it |
| M-us | Unstructured, KR against Satterthwaite | evaluated (10) | Satterthwaite 0.076, KR 0.059 |

### 3.4 Manuscript 06 (component decomposition)

Table 16: Manuscript 06 component-decomposition strategies C1 to C4 with evidence

| ID | Specification | Status | Evidence |
|---|---|---|---|
| C1 | One-component (`bm:Dbc`) | evaluated | unbiased if the biomarker is unrelated to PB and TV; size 0.031 to 0.056 |
| C2 | Phase-augmented: phase, phase by drug, phase by biomarker by drug | evaluated | no bias reduction; less power; size 0.038 to 0.049 |

| ID | Specification | Status | Evidence |
|---|---|---|---|
| C3 | Belief-covariate decomposition | evaluated | recovers the pharmacological slope only on a balanced-placebo design |
| C4 | Blinded-stratum contrast; Gompertz-basis decomposition | evaluated | fail on the Hybrid design |

### 3.5 Summary measures (`docs/37`)

Table 17: Summary-measures statistics S1 to S3 with status and evidence

| ID | Specification | Status | Evidence |
|---|---|---|---|
| S1 | E9: post-baseline on-minus-off contrast, OLS on biomarker | evaluated (closed form and Monte Carlo) | exact closed form; validated over 566 cell-variants |
| S2 | E9b: baseline counted as off drug | evaluated | mirrors O1's baseline coding; 15% to 16% noisier than E9 without carryover (`docs/41`) |
| S3 | Stratified E9: path-specific intercepts | closed form only | removes the CO period-effect loss |

### 3.6 This discussion (`docs/40` to `docs/42`; published CS construct)

Table 18: Approaches D1 to D7 from docs/40 to docs/42 with evidence on size

| ID | Specification | Status | Evidence on size |
|---|---|---|---|
| D1 | O1 with `ns(t, 3)` | evaluated | Hybrid t½ 1: 0.073 (O1 0.058); recurs in `docs/45` (0.070), absent under CR2 |
| D2 | O1 with visit factor | evaluated | Hybrid t½ 1: 0.080; recurs in `docs/45` (0.072), absent under CR2 |
| D3 | O1 plus fixed path effect | evaluated | power changes by at most 0.001 |
| D4 | O1 on a single path against the mixture | evaluated | no material difference in Hybrid |

| ID | Specification | Status | Evidence on size |
|---|---|---|---|
| D5 | Randomized (multinomial or permuted-block) path allocation | proposed | |
| D6 | Random intercept, post-baseline shift and expectancy coefficient, `varIdent` by phase | proposed; partly evaluated as 46-G1, 46-G2 (fixed, not random, expectancy coefficient) | 46-G1 pooled 0.049, 46-G2 0.047 |
| D7 | Baseline-by-drug moderator | proposed | |

### 3.7 The covariance study: main run (published CS and configuration C)

All `mmrm`, centered biomarker, binary `Db`. Fitted to the same trials: 16 cells, 250 alternative and 500 null replicates.

Table 19: Covariance study main-run approaches V1 to V7 with size ranges

| ID | Coding | Covariance | Time | Test | Size range |
|---|---|---|---|---|---|
| V1 | row | `cs` | linear, spline, visit | Satterthwaite | CS 0.036 to 0.072; configuration C, CO 0.140 to 0.158 |
| V2 | row | `csh` | linear, spline, visit | Satterthwaite | as V1 (CO, configuration C: 0.138 to 0.164) |
| V3 | row | `us` | linear, spline, visit | Satterthwaite | 0.072 to 0.104 |
| V4 | row | `sp_exp` | linear, spline, visit | Satterthwaite | 0.026 to 0.062; low power under CS |

| ID | Coding | Covariance | Time | Test | Size range |
|---|---|---|---|---|---|
| V5 | row | `us` | linear, visit | Kenward-Roger | 0.046 to 0.072 |
| V6 | covariate | `cs` | linear, spline, visit | Satterthwaite | CS 0.034 to 0.064; configuration C, CO 0.146 to 0.164 |
| V7 | covariate | `us` | linear, spline, visit | Satterthwaite | 0.072 to 0.096 |

### 3.8 The covariance study: follow-up run (completed; same trials)

The run was in progress when this catalog was written. The last column was added on 2026-10-07 from `docs/45` (Section 4.1 mean sizes over the eight null cells, linear time; Section 6 for F7).

Table 20: Covariance study follow-up approaches F1 to F7, their questions and answers

| ID | Coding | Covariance | Time | Test | Question | Answer (`docs/45`) |
|---|---|---|---|---|---|---|
| F1 | covariate | `us` | linear, spline, visit | Kenward-Roger | is the covariate coding correctly sized? | slightly liberal, 0.059 |
| F2 | covariate | `cs` | linear, visit | Kenward-Roger | does KR change a parsimonious model? | no; liberal, 0.077 |
| F3 | row | `us` | spline | Kenward-Roger | completes V5 | same class as linear time, within 0.005 |
| F4 | row | `cs`, `csh` | linear, visit | Kenward-Roger | can KR repair a misspecified covariance? | no; 0.079 and 0.073 |

| ID | Coding | Covariance | Time | Test | Question | Answer (`docs/45`) |
|---|---|---|---|---|---|---|
| F5 | row | `cs`, `csh`, `us`, `sp_exp` | linear, visit | CR2 | does CR2 hold size under every covariance? | nominal for `cs`, `csh`, `sp_exp` (0.048 to 0.054); liberal for `us` (0.086) |
| F6 | covariate | `cs`, `us` | linear, visit | CR2 | as F5, covariate coding | `cs` nominal (0.048); `us` liberal (0.080) |
| F7 | row, separate baseline biomarker effect | `us` | visit | Kenward-Roger | does the shared biomarker effect explain the row coding's power? | yes: freed, power matches the covariate coding (0.780 against 0.792) |

### 3.9 The stress test (`docs/46`; added 2026-10-07)

All 14 analyses were fitted to the same simulated Hybrid trials: 41 cells (39 in the original working protocol, plus two prognostic-baseline cells, 40 and 41), 15 null cells of 1,000 replicates and 26 alternative cells of 500, 28,000 trials in all. The base construct is configuration C at $\rho = 0.7$, $N = 70$, $t_{1/2} = 0.5$ week. Every analysis has linear time and a centered biomarker (`bmc`), and uses $D_{bc,it}$ at $t^a_{1/2} = 0.5$ week, except 46-R (binary $D_{it}$) and the AIC arms 46-DS, 46-DA and 46-F1A (half-life chosen over $\{0, 0.25, 0.5, 1, 2\}$). IDs carry the 46- prefix of the Identifiers note above.

Table 21: Stress-test analyses with null rejection rates and screen outcomes

| ID | Specification | Pooled size | Worst non-prognostic cell | Prognostic cells (BL, PB, TV constant, TV growing) | Outcome |
|---|---|---|---|---|---|
| 46-S | random intercept, `corCAR1`, CR2, baseline row | 0.048 | 0.059 | 0.050, 0.100, 0.099, 0.049 | survives; flagged |

| ID | Specification | Pooled size | Worst non-prognostic cell | Prognostic cells (BL, PB, TV constant, TV growing) | Outcome |
|---|---|---|---|---|---|
| 46-A | `us`, Kenward-Roger, baseline row (`docs/45` primary) | 0.057 | 0.087 ($N = 35$) | 0.051, 0.222, 0.416, 0.195 | fails size |
| 46-C1 | 46-S with `varIdent` by phase, CR2 | 0.053 | 0.069 | 0.054, 0.205, 0.382, 0.173 | survives; flagged |
| 46-C2 | 46-C1, model-based | 0.052 | 0.066 | 0.050, 0.193, 0.393, 0.172 | survives; flagged |
| 46-DS | 46-S, $t^a_{1/2}$ chosen by AIC | 0.053 | 0.068 | 0.055, 0.112, 0.115, 0.058 | survives; flagged |
| 46-DA | 46-A, $t^a_{1/2}$ chosen by AIC | 0.065 | 0.103 ($N = 35$) | 0.064, 0.256, 0.509, 0.245 | fails size |
| 46-ES | 46-S with $\beta_B$ freed at baseline | 0.048 | 0.059 | 0.050, 0.049, 0.049, 0.063 | survives; robust |
| 46-EA | 46-A with $\beta_B$ freed at baseline | 0.052 | 0.070 | 0.066, 0.050, 0.049, 0.057 | survives; robust |
| 46-F1 | 46-S plus `De + bmc:De` | 0.051 | 0.062 | 0.040, 0.040, 0.047, 0.046 | survives; robust; **selected** |
| 46-F2 | 46-S plus `bmc:t` | 0.049 | 0.059 | 0.048, 0.098, 0.090, 0.048 | survives; flagged |
| 46-R | published reference: random intercept, model-based, binary $D_{it}$ | 0.069 | 0.081 | 0.075, 0.159, 0.107, 0.053 | fails size |
| 46-G1 | 46-F1 mean model, phase-specific residual variances, CR2 | 0.049 | 0.060 | 0.042, 0.042, 0.055, 0.044 | survives; robust |
| 46-G2 | 46-G1, model-based | 0.047 | 0.066 | 0.046, 0.045, 0.062, 0.043 | survives; robust |
| 46-F1A | 46-F1, $t^a_{1/2}$ chosen by AIC | 0.057 | 0.073 ($t_{1/2} = 1$) | 0.045, 0.047, 0.059, 0.052 | fails size |

Pooled size is over the 11 non-prognostic null cells (11,000 trials); the nominal band is 0.0458 to 0.0542 (2 MCSE) with no cell above 0.075. *Flagged* means the analysis fails at least one prognostic cell; *robust* means it is nominal in all four. All 14 pass the monotonicity screen; the symmetry screen was dropped in Amendment 1 of `docs/46`, because power is symmetric in the sign of $c_{bm}$ by construction. Maximin shortfall of size-adjusted power over the non-prognostic alternative cells: 46-C1 0.034, 46-C2 0.040, 46-S 0.144, 46-F2 0.160, 46-DS 0.182, 46-G2 0.232, 46-F1 0.242, 46-G1 0.252, 46-ES 0.278, 46-EA 0.318. 46-C1 has the smallest maximin but fails the prognostic cells. Among the robust survivors 46-G2 and 46-F1 tie within 0.02; neither reaches power 0.80 anywhere on the power-curve grid, so the tie goes to the analysis with fewer covariance parameters, 46-F1. Failed fits were 0 to 12 per analysis out of 28,000 (at most 0.6% in a cell; verified from the `fails` column of `summary-full*.csv`). The working protocol counted failed fits as non-rejections; the summaries exclude them, which changes no rate by more than 0.0025. (Verified against `analysis/data/quick-sim/carryover-closed-form/strawman-stress/decision-*-ds-nosym-all14.csv`; decision rule rerun with all 14 analyses, 2026-10-07.)

## 4. What is known about size

**Main conclusions:**

1. **The covariance and its small-sample correction dominate.** Changing the exposure coding (02), the time adjustment (`docs/40`) or the baseline coding (covariance study) moves the Type I error by at most about 0.02. Changing the working covariance or the correction moves it by up to 0.10.
2. **A parsimonious covariance that is wrong fails in either direction.** The published random intercept is anticonservative under AR(1) data in CO (0.14 to 0.16). The random intercept with `corCAR1` is conservative in the package construct (0.029 to 0.039). CR2 restores the latter. Kenward-Roger does not, being inert on parsimonious structures (10).
3. **An unstructured covariance needs Kenward-Roger.** With Satterthwaite alone it is anticonservative in every construct tested (0.072 to 0.104 in the covariance study, 0.076 in 10). Kenward-Roger brings it to nominal. *Corrected (2026-10-07):* Kenward-Roger brings it close to nominal but not to it: slightly liberal over the covariance study's null cells (mean 0.058, `docs/45`), and failing the stress test's size criterion (46-A: pooled 0.057, 0.087 at $N = 35$).
4. **Sandwich estimators need small-sample corrections.** Naive GEE is anticonservative. Mancl-DeRouen and CR2 restore the size.
5. **A residual size problem remains for flexible time adjustments in Hybrid under carryover** (about 0.07). It persists under every covariance tested, including `us` with Kenward-Roger, and its cause is unexplained. *Corrected (2026-10-07):* the follow-up run tested CR2, under

which the excess is absent (0.058 to 0.060 in that cell); under model-based and Kenward-Roger inference it is real, recurring in a second run at 0.068 to 0.072 (`docs/45`, Section 4.4, whose description of this evidence as weak is itself corrected in `docs/42`). Its cause remains unexplained.

6. **The stress test separates the prognostic-robust analyses (added 2026-10-07).** Every analysis that neither frees $\beta_B$ at baseline nor adds the expectancy term fails at least one cell in which the biomarker is prognostic for $TV$ or $PB$ (null rejection up to 0.509). Freeing $\beta_B$ at baseline (46-ES, 46-EA) or adding `De + bmc:De` (46-F1, 46-G1, 46-G2) gives nominal size in every prognostic cell; `bmc:t` (46-F2) repairs only the cell with a growing natural-course association (Section 3.9).

**Construct dependence.** Results come from three constructs: published CS, configuration C, and the package construct. Approaches have not all been tested in all three. The parsimonious covariances are the most construct-dependent: the random intercept is well calibrated under CS data (10 M1; covariance study V1 under CS) and fails under AR(1) data.

## 5. Approaches correctly sized in every cell tested

Table 22: Approaches correctly sized in every cell tested, with relative power

| Approach | Constructs | Size range | Power relative to the published analysis |
|---|---|---|---|
| Row, `us`, Kenward-Roger, linear or visit (V5) | published CS, configuration C | 0.046 to 0.072 | similar under CS; +0.10 to +0.12 in Hybrid under configuration C |
| Random intercept plus `corCAR1`, CR2 (02 G6 to G9; 10) | package | 0.045 to 0.049 | slightly higher than model-based (recovers its conservatism) |
| GEE, Mancl-DeRouen (T5) | package | nominal | modest power cost against the mixed model (08) |
| `sp_exp`, Satterthwaite (V4) | published CS, configuration C | 0.026 to 0.062 | much lower under CS (0.31 to 0.52 against about 0.70) |

V5's upper value of 0.072 occurs with the visit-factor time adjustment in Hybrid at t½ = 1 (published CS), the residual problem of Section 4, point 5. With linear time V5 stays within 0.052 to 0.066.

*Corrected (2026-10-07).* Two rows of this table do not meet the catalog's own criterion. V5 is slightly liberal (mean 0.058, `docs/45`) and its stress-test counterpart 46-A, the same model with $D_{bc,it}$ at 0.5 week, fails the size criterion (pooled 0.057; 0.087 at $N = 35$) and all three original prognostic cells (0.195 to 0.416). It is therefore not recommended, and `docs/45`'s recommendation of it as the primary analysis is superseded by `docs/46` and `docs/49`. V4 (`sp_exp`, Satterthwaite) is conservative (mean 0.042, minimum 0.028; `docs/45`). The analyses now known to be correctly sized and robust in every stress-test cell, prognostic cells included, are 46-ES, 46-EA, 46-F1, 46-G1 and 46-G2 (Section 3.9), all in the configuration C and compound-symmetry constructs of `docs/46`; 46-F1 is the selected analysis.

## 6. Gaps and the evaluation in progress

**Being answered by the follow-up run (F1 to F7):** Kenward-Roger and CR2 for the baseline-covariate coding; whether CR2 rescues the misspecified parsimonious covariances under AR(1) data; whether Kenward-Roger does (expected not); and the mechanism behind the row coding's power under AR(1). *Answered (2026-10-07)* in `docs/45`; see the last column of Section 3.8.

**Not yet scheduled** (status of each item added 2026-10-07):

1. The cause of the flexible-time size excess in Hybrid under carryover (Section 4, point 5). *Open;* CR2 is known to remove it.
2. The random-effects structure implied by the published construct, with phase-specific variance (D6). *Partly done:* 46-G1 and 46-G2 (fixed expectancy term, phase-specific residual variances).
3. A random slope on time (O3) and the expectancy term (O2) by simulation, not only in the population-mean limit. *Partly done:* the expectancy term with `bmc:De` under CR2 is 46-F1; O3 is open.
4. GEE with Mancl-DeRouen and the summary-measures statistic in the published CS and configuration C constructs, for a direct comparison with V5. *Open.*
5. Exposure-weighted coding (`Dbc`) crossed with the covariance choices. Manuscript 02 evaluated it only under `corCAR1`. *Done:* `docs/45` exposure pass (`us-KR`, `us-CR2`, `RI+CAR1-CR2`; no change in size) and every stress-test analysis except 46-R.
6. Randomized path allocation (D5). *Open.*
7. Power at matched size, for every approach found correctly sized. *Done* for the analyses of `docs/45` and `docs/46`.
8. Bayesian hierarchical models and randomization tests, named in the literature reviews of 02 and 08 and not evaluated. *Open.*

**Suggested core comparison.** Once the follow-up run is in, a single paired study in the three constructs could cover:

- the approaches correctly sized so far (V5, CR2 variants, GEE with Mancl-DeRouen);
- the published analysis as reference (O1);
- crossed with the exposure coding (`Db` against `Dbc`) and the baseline coding.

That would give one table of size and power from which a primary analysis for a Hybrid trial could be chosen. *Status (2026-10-07):* the exploratory stress test of `docs/46` is that study for the Hybrid design, in configuration C and compound symmetry but not the package construct, and without GEE or the summary-measures statistic. It selected 46-F1 (Section 3.9; `docs/49`).

## 7. Sources

- `orig`: `github.com/rchendrickson/pmsimstats`, commit `58b32a9`, `R/lme_analysis.R` and vignette 1 (inspected).
- Manuscripts: `analysis/report/02-carryover-sensitivity/`, `06-component-decomposition/`, `08-test-procedure-design-sensitivity/`, `10-interaction-test-calibration/` (read for their specifications and headline results; the numbers quoted are those reported in the manuscripts).
- Project documents: `docs/26`, `docs/31`, `docs/36`, `docs/37`, `docs/40`, `docs/41`, `docs/42`; added 2026-10-07: `docs/45`, `docs/46`, `docs/48`, `docs/49`.
- Covariance study: `analysis/scripts/quick-sim/carryover-closed-form/08-covariance-study.R`; results in `analysis/data/quick-sim/carryover-closed-form/covariance-study/summary-alt250-n` (verified); follow-up completed and reported in `docs/45`.
- Stress test: `analysis/scripts/quick-sim/carryover-closed-form/11-strawman-stress-test.R` and `12-strawman-stress-decision.R`; results in `analysis/data/quick-sim/carryover-closed-form/` (`decision-*-ds-nosym-all14.csv`, `summary-full*.csv`; verified).